Motion-Only Capsize Warning on an Audited Synthetic Benchmark: Ranking Skill Without Threshold Transfer

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Abstract

Can the measured roll motion of a ship warn of capsize soon enough to act, at a false-alarm cost a crew could accept? We study this question on a validated synthetic benchmark: a one-degree-of-freedom nonlinear roll model, three restoring-curve families representing distinct ways a ship loses stability, and 58,500 seeded trajectories under JONSWAP forcing. Five warning methods compete under one protocol in which every operating threshold is chosen on calibration data and spent once on test data: a small convolutional network, classical variance and autocorrelation trend statistics, a likelihood-ratio detector, a closed-form critical-roll-rate margin, and a phase-space neighbor-count score. Four findings emerge. (a) Ranking skill is real at every forcing bandwidth tested; the fixed-window primary network separates dangerous from ordinary intervals with window AUC 0.883 to 0.913. (b) Inside an established severe regime, no method predicts which encounter will be fatal; all score near chance. (c) Most nominally false alarms, 75 to 88 percent, coincide with genuine critical wave groups that the vessel survived. (d) No method's threshold transfers across failure modes at fixed sensitivity. We then operate the split-time upcrossing decomposition as an onboard rate estimator, in online and offline-calibrated forms; both fail preregistered calibration tests for a reason we localize. The decomposition itself is sound: given the empirically measured probability that a threshold crossing is terminal, it reproduces realized counts within 3 in 71. But the closed-form criticality model overestimates the lethality of a crossing by roughly a factor of four, and in 92 percent of capsizes the crossing that matters arrives less than ten seconds before the event. An audit during the study found our own first results inflated by test-set threshold selection; every number reported here is the corrected value. The picture that survives is simple: motion reveals the slowly building vulnerability, and only the sea knows the moment.

Keywords: capsize, intact stability, real-time warning, split-time method, preregistered evaluation, synthetic benchmark.

1. Introduction

Begin from first principles. A ship in beam seas is a lightly damped nonlinear oscillator: waves push, the righting arm restores, and capsize is the escape of that oscillator over the rim of its potential well. Two different questions hide inside "can we warn?" One asks whether the well is becoming shallow — vulnerability, a slow process. The other asks when a particular group of waves will push the ship over — the encounter, a fast one. Every result in this paper answers one of those two questions, and the paper's central finding is that motion measurement answers only the first.

Interest in warning of capsize from onboard motion measurement is at least two decades old. The Lyapunov-exponent programme of the mid-2000s (McCue and Troesch 2004, 2006) sought a precursor of capsize in the divergence of measured trajectories; the author's thesis in that programme (Story 2009) found empirically that the useful signal lay not in estimated exponents but in the loss of phase-space neighbors, and reported a neighbor-count warning flag with encouraging lead times and an honestly documented false-alarm problem. The thread was not taken up. The signal-based detectors of Galeazzi et al. (2013, 2015) for parametric roll, with designed false-alarm probabilities and full-scale blind validation, remain the only fielded success of motion-based stability-failure detection, and the field's general machinery moved instead to offline probabilistic assessment (Belenky and Sevastianov 2007), of which the split-time method (Belenky, Weems and Lin 2016; Belenky et al. 2024) is the principal example, and to operational guidance derived from precomputed simulation under the second-generation intact stability criteria (Bačkalov et al. 2016; IMO 2020; Petacco and Gualeni 2020; Shigunov 2023).

The relationship of the present work to the split-time method deserves statement at the outset, since that method supplies both the strongest available machinery and the sharpest open question. Belenky et al. (2024) compute the probability of capsizing offline. Their central object is an upcrossing: the roll angle passing outward through a fixed intermediate threshold, here the angle of maximum righting arm. The metric of danger, a critical roll rate at those upcrossings, is evaluated along simulated histories, its tail is extrapolated with a physics-informed exponential form, and the estimate is validated by confidence-interval capture against direct counting. Three transfers from that programme are attempted here. The metric itself is operated, to our knowledge for the first time, as a continuously evaluated onboard alarm statistic computed from measured state alone, and is benchmarked against learned and statistical competitors on equal terms. The statistical discipline of the split-time programme, declustering, interval estimates on every rate, and validation against direct counting, is transplanted to warning evaluation, where it is presently absent. And the boundary of the transfer is itself measured: the engineering-fidelity form of the metric requires re-simulation with the incident wave realization known, and the present results quantify operationally what that requirement conceals, namely that the information carried by the future waves is absent from measured motion precisely where event timing is decided.

The present study returns to the motion-only question with tools that did not exist in 2009 and with an evaluation protocol designed against the failure modes that make warning research easy to overstate. A warning method may rank dangerous intervals well yet possess no usable threshold; a threshold may fail to transfer across failure modes or sea states; apparent skill may arise from protocol time, truncated outcome windows, or the use of future data during normalization; and, because capsizes are rare, an aggregate score conceals all of these. Each failure mode is tested separately here. The study also reports, deliberately, an audit of its own methodology: the operating points first obtained were later found to have been selected on test outcomes, an error we believe is widespread in the published literature on learned motion prediction, and the corrected results quantify its size.

Three questions are addressed. First, does motion history rank capsize risk under forcing broad enough that classical critical-slowing-down indicators fail? Second, do the resulting operating thresholds transfer across stability-failure modes at fixed sensitivity? Third, what do the false alarms of such systems consist of?

2. The synthetic benchmark

The benchmark is the simplest ship that can capsize (Figure 1): a one-degree-of-freedom nonlinear roll model, validated against analytic limits, generating seeded trajectories under JONSWAP forcing (Hasselmann et al. 1973). Three restoring families represent failure archetypes: softening restoring (dead-ship escape), time-varying stiffness (parametric roll), and biased restoring with asymmetric escape angles (damage, steady heel), in the scope of established one-degree-of-freedom roll comparisons (Bulian and Francescutto 2004, 2011). Following-sea failure modes, broaching foremost (Umeda et al. 2016), are outside the benchmark's scope, a limitation the transfer results of Section 5 should be read against. Forcing is synthesized spectrally with deterministic amplitudes and seeded random phases (Shinozuka and Deodatis 1991); integration is fixed-step Runge-Kutta at no fewer than 40 steps per natural period; capsize is an absorbing state and post-event samples are excluded from every window.

Validation gates precede all warning experiments: recovered significant wave height within 2 percent; linear-limit response variance within 6 percent of the spectral calculation; the principal Mathieu boundary within 10 percent of the averaging result; and the simulated harmonic capsize boundary at or above the heteroclinic Melnikov threshold (Falzarano, Shaw and Troesch 1992) with the correct frequency shape, the bound being treated as a necessary condition only. No validation test is skipped in the acceptance run.

The reference data comprise 58,500 trajectories of 600 seconds at a four-second natural period: stationary campaigns per family, stiffness-ramp campaigns representing slow stability erosion, one sea-state step campaign, rare-event evaluation campaigns at 0.95 to 2.0 percent capsize rates, and five forcing-bandwidth campaigns in which JONSWAP peak enhancement is varied from 1.0 to 30 with terminal ramp severity retuned per bandwidth so that outcome rates remain in a common band; the last decision prevents severity from acting as a hidden bandwidth axis. Disjoint seed blocks separate training, calibration, test, and guarded holdout data.

Supervised windows use 60 natural periods of causally normalized roll and roll-rate history, a 50-period outcome horizon, and an exclusion band for near misses; a future-only leakage probe verifies the entire feature path against a deliberately leaky control. The label discipline responds directly to critiques of early-warning-signal studies in other fields (Boettiger and Hastings 2012; Dakos et al. 2012).

3. Five warning methods

All five methods consume identical windows and emit a scalar score which is thresholded into alarm episodes (three consecutive crossings to open; refractory closure). No method receives wave elevation, spectrum, family identity, or sea-state input.

(a) A temporal convolutional network of 2,969 parameters trained on the horizon labels, in the spirit of learned early-warning classifiers (Bury et al. 2021); neural roll prediction has in-community precedent (Míguez González et al. 2023). (b) Rolling variance and lag-one autocorrelation with Kendall trend statistics, the classical critical-slowing-down indicators (Dakos et al. 2012). (c) A roll-power adaptation of the double-Weibull generalized likelihood-ratio detector of Galeazzi et al. (2013, 2015). (d) The split-time critical roll rate (Belenky et al. 2024), computed in closed form from the known restoring parameters and the measured state, with the separatrix line extrapolated between threshold crossings; to our knowledge this quantity

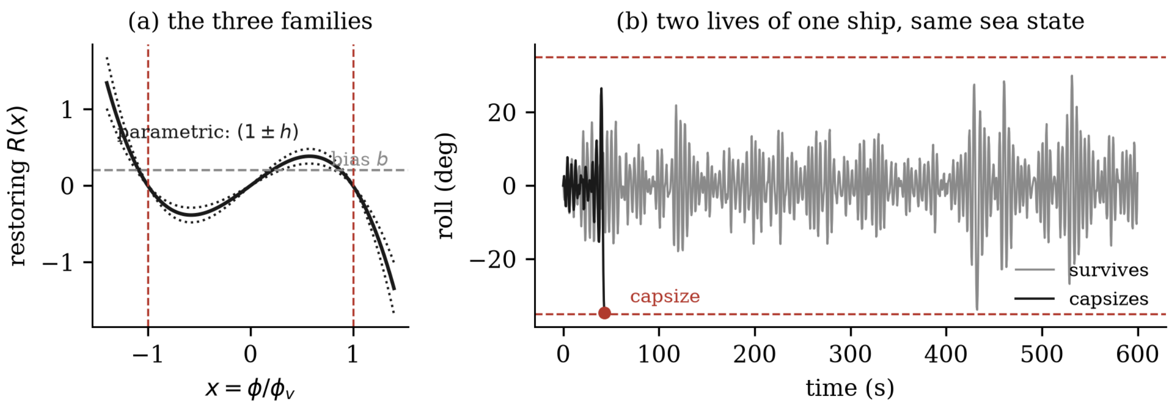

Figure 1: The benchmark. (a) The three restoring families: the softening curve with its escape angles (dashed red), the constant bias moment that shifts the biased family's equilibria, and the parametric family's stiffness modulation envelope (dotted). (b) Two 600-second lives of the same ship in the same sea state, different seeded seas: the capsizing life ends during a wave group in its first minute, while its twin survives larger excursions for the full record. The encounter, not the general severity, decides the moment.

has not previously been operated as a continuously evaluated online alarm statistic, and it functions here as a zero-training physics baseline. (e) The neighbor-count score of Story (2009), reimplemented faithfully in the (roll, roll-rate) plane under strictly causal normalization, with the thesis's fixed flag retained as one point on a full operating curve; phase-space identification of dangerous ship states has since been developed independently in the Spyrou school (Kontolefas and Spyrou 2020).

4. Evaluation protocol and the audit

Metrics are episode sensitivity, false episodes per exposure hour, and lead time, with trajectory-block bootstrap intervals and fixed campaign weights; exposure ends at capsize; an episode overlapping the pre-capsize horizon is event-associated. All operating controls are selected on calibration data, frozen, and evaluated once on test data.

The protocol was not followed from the outset, and the failure is reported as a result. The study's first operating points were obtained by scanning test-set metrics for the lowest false-episode rate at 90 percent sensitivity or better, which is threshold selection on test outcomes. For the network on the pooled rare-event campaigns, the test-selected report was 6.3 false episodes per hour; the superseded pre-v0.2 calibration-frozen value for the same detector on the same data was 15.5 per hour at 92.4 percent sensitivity. A factor of approximately 2.5 in reported alarm cost was therefore attributable to the selection procedure rather than the detector, at a sensitivity target of 90 percent and capsize prevalences of 1 to 2 percent. One guarded holdout evaluation was additionally invalidated by threshold selection on the holdout labels, and its artifacts are retained as historical records only. The corrected protocol is enforced by the test suite rather than by convention (Figure 2). We suggest that reported operating points in the learned motion-prediction literature should be read with this mechanism in mind wherever

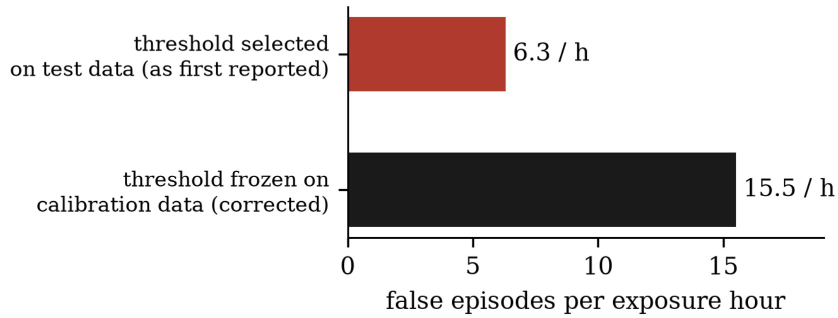

Figure 2: The audit's effect. The same detector on the same data, with only the provenance of the operating threshold changed: selecting the threshold on test outcomes understated the false-alarm cost by a factor of approximately 2.5.

threshold provenance is not stated.

5. Results

Ranking under bandwidth (D3). The fixed-window primary network's window AUC was 0.883 to 0.913 at every peak enhancement from 30 down to 1.0, while the classical trend indicators collapsed on ramped campaigns. Ranking skill from motion history survives broadband forcing. The predeclared operating-cost criterion at the broadband end (a 10 percent false-episode improvement) was missed at 7.6 percent (14.061442 versus 15.212539 false episodes per hour), and the operating-cost claim is reported as inconclusive: ranking skill and a deployable threshold are different assets (Figure 3).

Pooled operating points (D1). At calibration-frozen thresholds on the pooled rare-event campaigns, the fixed-window primary network attained 91.00 percent sensitivity at 13.409 false episodes per hour; the classical indicators reached 100 percent sensitivity only at a near-always-on threshold costing 21.4 per hour; the physics margin and the 2009 neighbor score occupied the same high-sensitivity, high-cost region. No method produced an operating point that a master would tolerate (Figure 4).

Threshold transfer (D2). The fixed-window CNN reached 99.15 percent, 90.83 percent, and 76.21 percent test sensitivity across the softening, parametric, and biased holdouts. Two rotations met 90 percent, but only near always-on alarm cost; the biased holdout failed. No deployable transfer was established.

Within-regime discrimination (D5). After the full wash-in, the fixed-window CNN's orientation-independent AUC was 0.556, below the 0.58 trigger; cumulative-online CNN reached 0.538 and classical scores remained near chance. Motion-only scores carried no usable information about which encounter would be terminal inside the established severe regime.

The character of false alarms (D4). Between 75 and 88 percent of nominally false episodes coincided with critical wave-group encounters, defined by an envelope criterion on the known synthetic seaway in the spirit of the critical-wave-groups literature (Themelis and Spyrou 2007; Anastopoulos et al. 2016), that the vessel survived. The figure is descriptive, since no matched

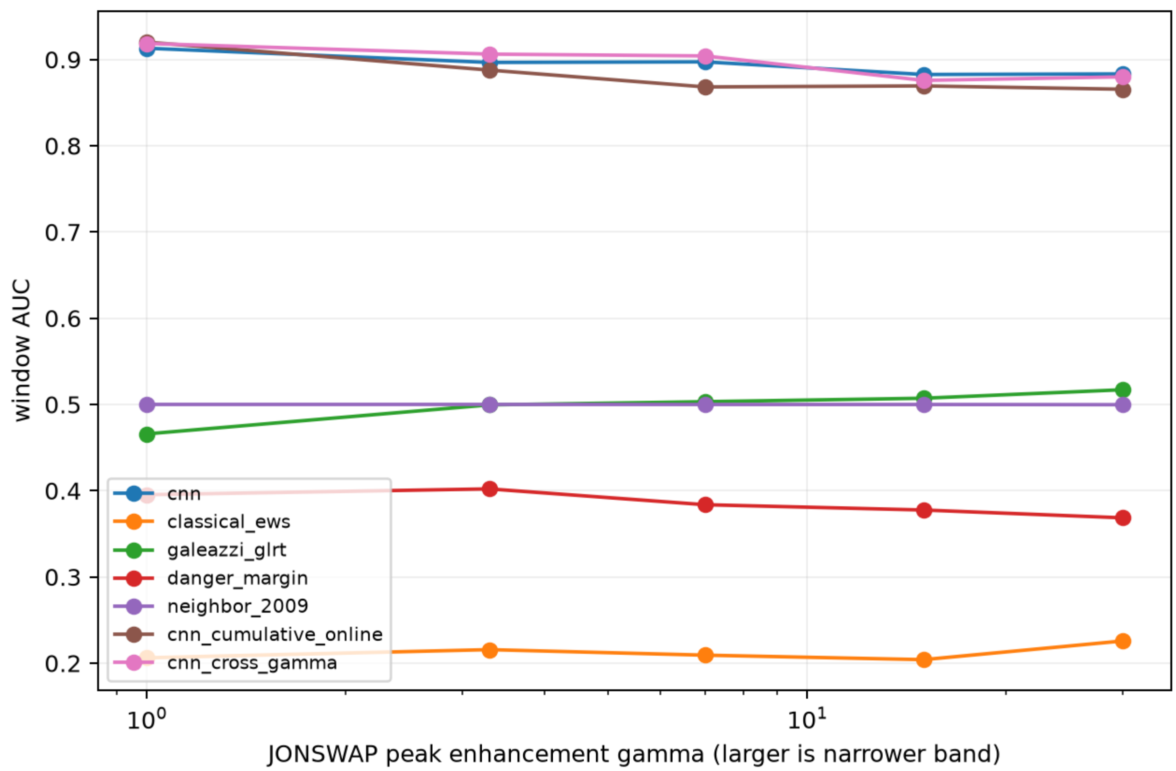

Figure 3: Ranking survives bandwidth. Window AUC on the five forcing-bandwidth campaigns, corrected record: the fixed-window primary network holds 0.883 to 0.913 from narrow-band forcing down to fully broadband, while the classical indicators and physics scores sit at or below chance on these ramped campaigns — the fixed-window scoring that is honest about information also strips the trends those baselines rely on.

null was tested and groups are common in a narrow-band sea; its qualitative content is that outcome-based false-positive accounting charges the detectors for hazard exposure the vessel happened to survive, the base-rate structure anticipated in Boettiger and Hastings (2012).

Reference comparators. On the v0.2 design-balanced window set, the exact-state independent-future restart reached AUC 0.850, fixed-window XGBoost 0.723, and fixed-window CNN 0.652; a protocol-time comparator also exceeded the network. Two qualifications apply: the restart comparators replace the realized forcing and are therefore reference points rather than bounds, since the temporally correlated seaway leaves encounter information in the motion history; and part of the network's advantage on natural test populations evidently derived from population structure rather than state inference. The headroom that exists lies in state inference, and simple tools claimed it first.

The split-time rate estimator (U1, H1). Three preregistered experiments operated the paper's ROM decomposition — capsize rate as upcrossing rate times the probability an upcrossing goes critical — as a continuously updated onboard estimate, each evaluated once on fresh test data by the split-time program's own confidence-interval-capture standard. The online-estimated form failed its capture criterion, and a predeclared comparison found the upcrossing

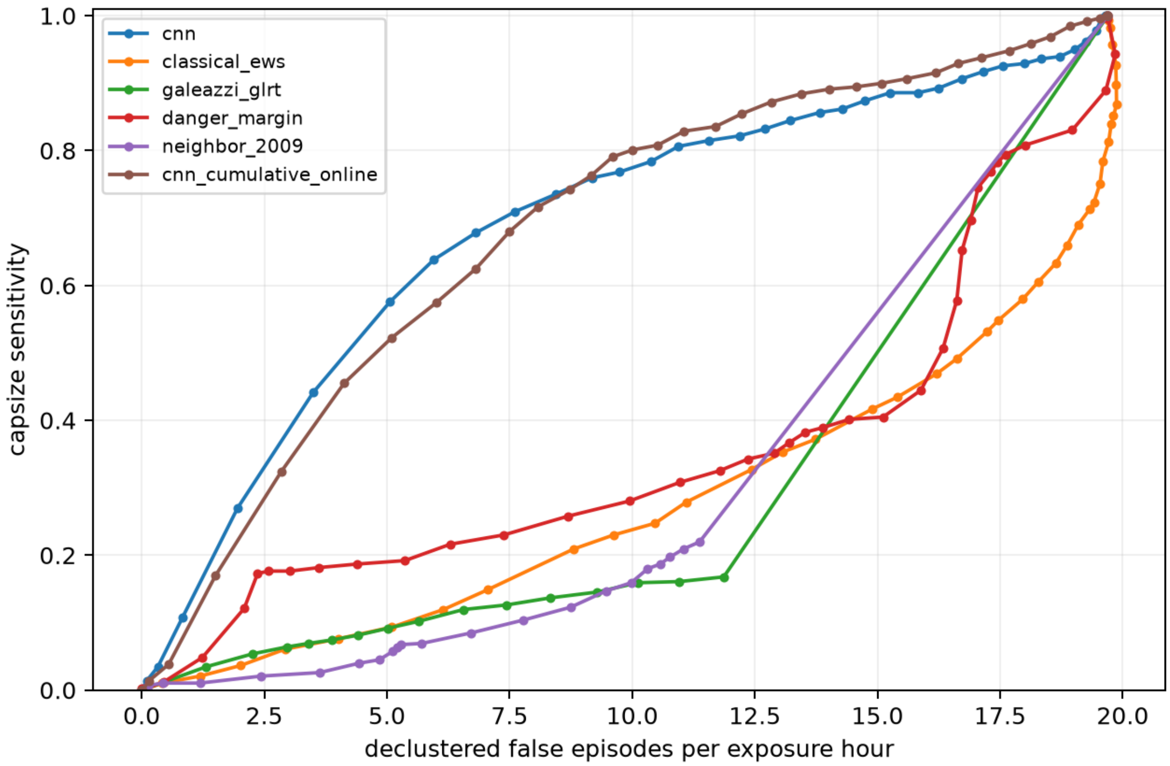

Figure 4: The pooled operating curves, corrected record: capsize sensitivity against declustered false episodes per exposure hour at calibration-frozen thresholds. The network variants dominate the low-cost region; every classical and physics score buys sensitivity only near the always-on corner. Even the best curve pays several false episodes per hour at 90 percent sensitivity.

rate alone more reliable than the full decomposition, so the corresponding kill criterion fired. A measured attribution then localized the error: with the empirical terminal-crossing probability substituted for the modeled one, the decomposition predicted 68 events against 71 realized, while the closed-form criticality model (the unforced piecewise-linear critical roll rate with an exponential tail) overpredicted by roughly a factor of four; a motion-derived estimate of the forced correction made the error larger, not smaller. A final hybrid, with the conditional calibrated offline from design-stage data and multiplied onboard by the measured crossing rate, also failed its capture and transfer criteria. A timing audit supplied the mechanism: in 92 percent of heralded capsizes (19 exceptions in 240) no emission interval separated the terminal crossing from the capsize, so the severity information the conditional carries becomes available only after the event has begun. Two methodological qualifications attach: fresh test slices of 30 to 60 expected events realized counts 21 to 37 percent from expectation, and the capture intervals carried parameter uncertainty without realization noise, so the capture criterion on slices of this size is severe even for a correctly calibrated estimator; the reliability errors (0.039848 for the hybrid, 0.042806 for the rate-only map, and 0.242303 for a variance-based map) are the fairer calibration comparison. All verdicts were preregistered and stand as recorded. Figure 5's middle panel shows the one encouraging pattern: the crossing rate alone, offline-mapped,

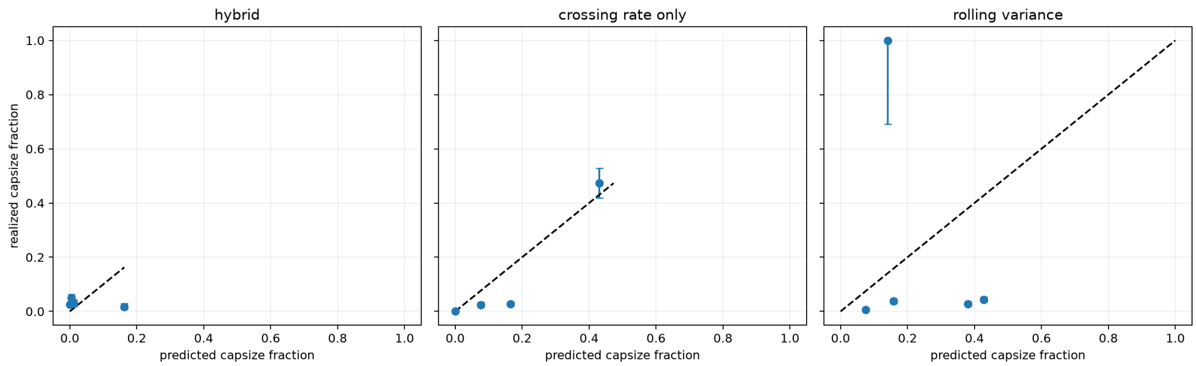

Figure 5: Reliability of the offline-calibrated hybrid and its comparators on fresh test data (H1): predicted capsize probability per bin against the realized fraction. The hybrid under-predicts throughout; the crossing-rate-only map tracks the diagonal; the variance map scatters. The observable factor of the split-time decomposition — the crossing rate — carries the calibration.

tracks the diagonal — the rate factor of the decomposition is the part that works.

A companion study of conformalized forecast intervals on the same benchmark (following Romano, Patterson and Candès 2019; Gibbs and Candès 2021) found stationary marginal coverage accurate to 0.75 percentage points in the mean, while every online adaptation scheme tested failed a predeclared joint rolling-coverage and alarm-cost criterion through the sea-state step. Those results are reported in the project record and not further discussed here.

6. Discussion

The results organize into a two-clock description. The slowly varying stability state, the eroding restoring that determines whether the vessel is in danger, is observable from motion history: ranking skill survives every forcing bandwidth, and the most informative engineered feature is in effect an online stiffness estimate, a quantity this community has long extracted from the roll period (Terada et al. 2016; Míguez González et al. 2017). The terminal encounter, the particular wave group that converts vulnerability into capsize, is not observable from motion history once the severe regime is established; the within-regime result is chance-level for learned, classical, and physical scores alike, consistent with the stochastic-Melnikov view that a deterministic separatrix loses its predictive meaning under random forcing (Frey and Simiu 1993). The practical reading is that motion-only systems should be understood, and evaluated, as vulnerability monitors rather than event predictors, a role compatible with the operational measures framework of the second-generation criteria (IMO 2020) rather than with an alarm bell.

Read against the split-time programme, the results locate both the value and the limit of operating its metric onboard. As a zero-training detector the closed-form critical roll rate was sensitivity-dominant and cost-competitive with every learned alternative, which is at once a tribute to the physics and a caution against claims for learned detectors that have not faced it as a baseline. Its one systematic failure, degradation on stability-erosion campaigns where the assumed restoring becomes stale, identifies the missing component for operational use: online estimation of the current restoring state, whose feasibility the comparator results demonstrate, the roll period alone recovering most of the available headroom. The within-regime chance result then explains, in information terms, why the engineering-fidelity metric must re-simulate with the waves known: what the motion perturbation method obtains from the realized future is not present in the measured past. The U1 and H1 experiments made that boundary quantitative from inside the method itself. The rate factor of the decomposition, an observable of the slowly varying stability state, survives as a usable vulnerability measure; the severity factor, which would time the event, is in 92 percent of capsizes not available until the event has begun. The two-clock description of Section 6 is therefore not an external criticism of the split-time programme but a property its own decomposition exhibits when operated online: the observable factor works, and the unobservable factor is precisely the one the offline method recovers by re-simulating the future.

The false-alarm finding bears on how such monitors should be judged. If three quarters or more of false episodes are honest responses to genuine hazard exposure, then false episodes per hour, computed against outcomes, systematically understates the value of the warning and overstates the cost. Metrics conditioned on hazard exposure, and alarm policies whose budgets are conditioned on regime, appear to us a more productive direction than further detector refinement.

Finally, the audit deserves a place in the discussion rather than an apology. The evaluation stack that produced the leaked operating points had predeclared kill criteria, confidence intervals on every rate, frozen data splits, and guarded holdouts, and it still admitted a factor of 2.5 of optimism through threshold provenance. The mechanism is not exotic and is not confined to this study. The community's evaluation practice for learned warning methods would benefit from treating threshold provenance as a reportable property of every operating point.

7. Conclusions

On a validated synthetic benchmark spanning three stability-failure archetypes, motion-only warning methods rank capsize risk well above chance at every forcing bandwidth tested, fail to transfer operating thresholds across failure modes, and discriminate event timing at chance inside an established severe regime; the majority of their false alarms coincide with real critical wave-group encounters. A test-selected evaluation had earlier made the leading detector appear two and one half times cheaper in false alarms than it is. The split-time experiments sharpen the same conclusion from inside the method: the upcrossing rate survives as a simple, usable vulnerability measure, while the severity term that would time the event arrives, in 92 percent of capsizes, after the event has begun. Further progress on the timing question appears to require information about the incident wave field rather than refinement of motion-only architectures; shipboard sea-state estimation is an established literature (Nielsen 2017), and the benchmark, with its frozen splits and audited protocol, is suited to that study without modification.

Acknowledgments

The benchmark code, campaign definitions, and the complete numerical record, including the pre-audit results retained as historical artifacts, are available from the author.

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